
accelerate
collaborate
innovate
communicate
palm
PalmSource
Conference 2000



Palm OS® Technology: New Handheld APIs

David Fedor
Developer Support Team Lead
Palm, Inc.

PalmSource
Conference 2000

Topics to Cover

- Direct Color
- Time Zones
- Attention Manager
- Exchange Manager and Libraries
- PDI Library and UDA routines
- Helper APIs
- Locale modules
- Changes to existing APIs, fixes, etc.

Direct Color

- OS support for direct color, up to 24 bits
  - (The converse is called indirect or indexed)
- Direct color means each pixel specifies actual RGB value; not through a palette
  - So 0x00ff00 is always bright green
- Old APIs still work just like in 3.5, and new APIs can be used on indirect screens
  - New ones can sometimes be much more efficient if the screen is in direct mode
- Check if trap exists; can be on 3.5
  - See if nonzero and != SysUnimplemented

Direct Color APIs

- WinSetForeColorRGB(RGBColorType *newRGB, RGBColorType *oldRGB);
  - Same for WinSetBackColorRGB( ) and WinSetTextColorRGB( )
  - Older APIs are probably a better choice, unless you really need the extra control
- WinGetPixelRGB(x, y, rgbP)
  - On direct systems, much faster than WinGetPixel since it doesn't need to do a palette lookup

Direct Color Objects

- Bitmap Families can have a direct member
  - Older OSs didn't know about direct, so make sure there's an indirect (indexed) family member
  - New compression algorithm: PackBits
- WinScreenAttributes( ) returns a struct with information about what the hardware supports
  - Whether direct color is supported
  - Amount of VRAM available
  - Number of bits per R/G/B component used by the LCD controller

Time Zones

- As wireless is more prevalent, knowing the user's time zone becomes important
- The OS is explicitly not doing:
  - Automatic daylight savings adjustments
  - Listing time zones for cities
  - Changing app data when the user goes to another time zone
- Old system prefs deprecated, since they were confusing, limited, rarely used, and had no UI
  - minutesWestOfGMT (which wasn't)
  - daylightSavings (which was an enum)

Time Zones Functionality

- Users manually set their time zone and DST
- Apps use the info if they want, in addition to the existing Time Manager APIs
- UI was removed from General Prefs, and added to a new Date & Time panel, and Setup
- OS internal resources ‘cnty’ and some ‘tSTR’ resources removed

Time Zones New APIs

- New system prefs:
  - timeZone (Int16, offset in minutes from GMT)
  - daylightSavingsAdjustment: Int16; usually 0 or 60
- TimTimeZoneToUTC( )
- TimUTCToTimeZone( )
- SelectTimeZone( )
- GetTimeZoneTriggerText( )

Attention Manager

- Applications can prevent indicator from flashing on their title bar
  - Call AttnIndicatorEnable(false) in form's initialization, then on again when form closes
  - If no title bar, or if the form is modal, the Form Manager won't draw any indicator
- Notification sent: sysNotifyGotUsersAttention
  - Use this hook for additional hardware/software

Getting Attention

- Primary function: AttnGetAttention( )
- Parameters include:
  - Insistent or subtle
  - Flags to prevent or force Sound, LED, Vibration, or everything
  - Nagging rate and # of repetitions
  - Callback function and user data for it

Your Attention Mgr Callback

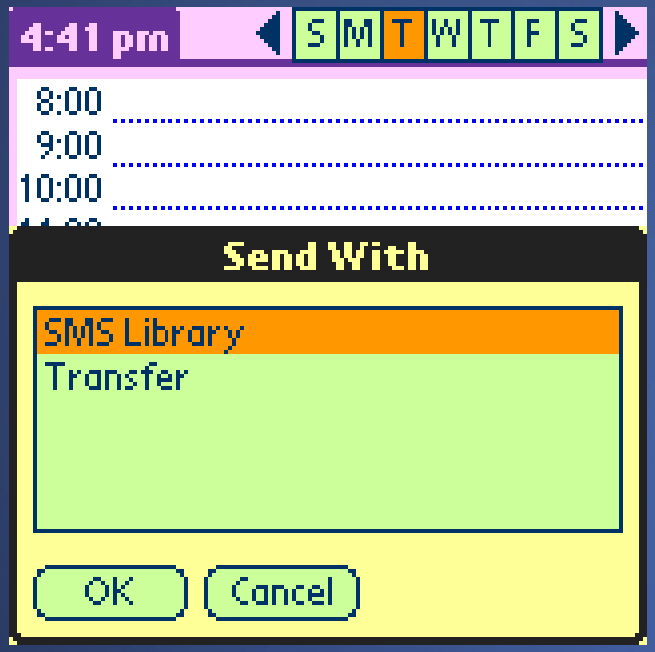
- Callback is called with commands:
  - Draw Detail view
  - Draw List view
  - Play Sound
  - Custom Effect
  - Go There
  - Got It (i.e., item is being dismissed)
  - Snooze
  - Iterate

Attention Mgr APIs

- AttnUpdate( )
  - Change display on screen
- AttnForgetIt( )
  - Remove an item from the list
  - Do this when user sees the object in your app
- AttnGetCounts( )
  - How many items are in the list (yours or everyone's)
- AttnIterate( )
  - Calls your callback for each active item

Exchange Manager Duties

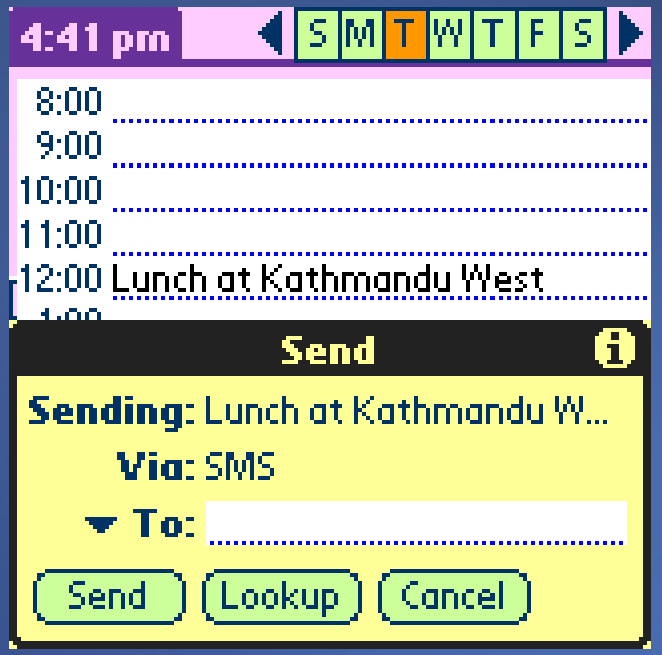
- Manage registry of libraries and applications
- Central API to send and receive data
- Central UI to choose between libraries



PalmSource Conference 2000

Exchange Library Duties

- Register for “Send” and/or another scheme
  - Also, underscore plus their own creator ID
- UI for addressing information, progress, errors
- Actually send or receive the data
- Can be an application
  - e.g., http and https are handled by Clipper



Making an Exchange Library

- Register for your scheme(s)
  - ExgRegisterData(… exgRegSchemeID, “http”, “Web URL”…)
- Implement each entrypoint in ExgLib.h
  - Mirrors of ExgMgr APIs: ExgLibPut, ExgLibSend...
- Or, for apps, just handle the GoToURL launch command
- Implement UI to set default app for a scheme

Receiving Objects

- When an Exchange Library receives an object, it gives it to the Exchange Manager with as much data as it can provide
  - MIME type
  - Filename with extension
  - Description
  - Etc.
- Registered application might show preview and dialog, or else the Exchange Manager will

Exchange Manager Advanced Use

- Prefix a scheme with “?” to bring up the UI for the user to pick between multiple libraries
  - Or don’t, to always go to the default one
  - “mailto:abcdefg@palm.com” versus “?mailto:abcdefg@palm.com”
- Specify a library’s creator ID as the scheme, to send directly to it
  - E.g., “_STRT” for the starter application
  - Or “_beam” or “_sms”

Multipart Objects

- Exchange Manager now allows multiple objects to be sent in a single transaction
  - Great for applications and required libraries
  - Call ExgPut multiple times
- Data Manager defines “bundled” databases, and the Launcher will beam them using this mechanism
- Incoming objects can have attachments. The API allows objects to be “unwrapped” and delivered at the right time

PDI Library

- Helps you read and write vObjects
  - vCard, vCal, vMessage, vNote, etc.
- Doesn't understand the data itself, nor the PIM record formats; that's up to the application
- Parses or properly formats the properties and parameters according to spec
- Handles the many formats for same data:
  - TEL;WORK;PREF:0123456789
  - TEL;TYPE=WORK;TYPE=PREF:0123456789
  - TEL;TYPE=WORK,PREF:0123456789
  - TEL;TYPE= "PREF", "WORK":0123456789

UDA Routines

- UDA routines are a framework to hook together the Exchange Manager, PDI Library, and applications
  - Actually a general-purpose mechanism
- Similar in concept to Unix piping
- You define a Reader, Writer, or Filter, and they get hooked together through callbacks
- Allows data to be sent to/from anywhere, without special knowledge
- Routines are in your app, not (necessarily) in the OS

Using PDI and UDA Together: AddrSendCategory, Step 1

- Step 1: AddrSendCategory creates Exchange Manager socket and calls ExgPut to begin the transaction
  - Exchange Library is chosen and is “underneath” the socket



Using PDI and UDA Together: AddrSendCategory, Step 2

- Step 2: AddrSendCategory creates UDAExchangeWriter object with that exgSocket
- Then calls AddrSendCategoryTryCatch
  - (To simplify cleanup in case of error)



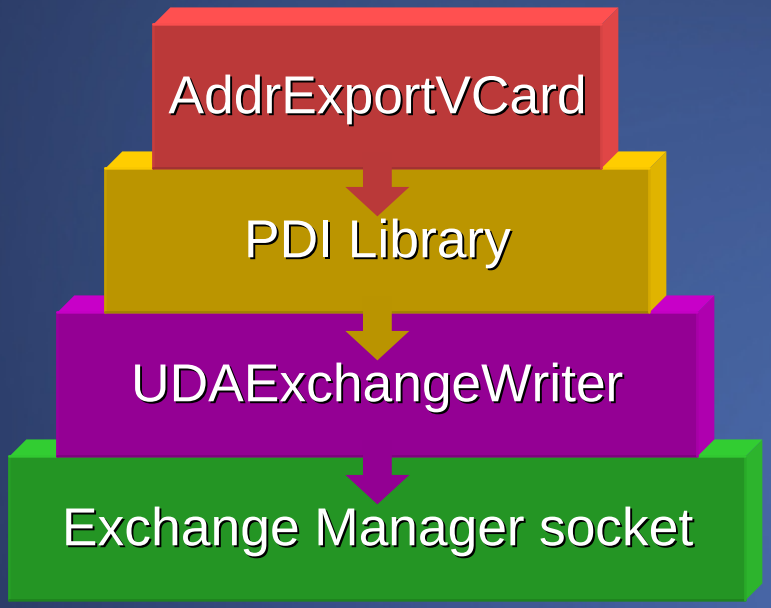
Using PDI and UDA Together: AddrSendCategory, Step 3

- Step 3: AddrSendCategoryTryCatch initializes PDI Library, providing output UDA object



Using PDI and UDA Together: AddrSendCategory, Step 4

- Step 4: AddrSendCategoryTryCatch calls AddrExportVCard in a loop, which calls the PDI library to create each object



Helper APIs

- A mechanism using the Notification Manager, for applications to find and use services provided by another application, without knowing which application is providing the service
- For well-defined services
  - Voice Phone call
  - Send an SMS message
  - Send a mail message (i.e., “mailto:”)
  - Send a Fax
  - … Others as mutually defined

Helper APIs

- Easier to code than Exchange Libs, but provides no UI mechanism nor system default selection nor handling of incoming data, and both sides must know how the service works
- Send notifications: Enumerate, Validate, Execute
- All helpers get all notifications; must abide by protocol and only take action if appropriate

Locale Modules

- Modularize the code and data which constitute a localization
- Support means for others to create new locales
  - New languages
  - New character encodings
  - New country settings
- No more complex language hacks which break for every new OS release!

Locale Modules

- Mostly all “under the hood”
- Data and tables and code to drive the Text Manager
  - Sorting, searching, word wrapping…
- Fonts, keyboard layouts…
- FEPs for text input

What Do Locales Mean for Me?

- Greater possibility of encountering a new character encoding, language, etc.
- Use the Text Manager, and don't look at the contents of a string
- Use the Locale Manager for info on the active language, country, time zone, number format…
- Consider translating your application
  - Overlays make it fairly easy

Various Improvements

- Soft-reset app is specifiable: prefDefaultAppCreator
- Hit-testing on form objects now happens in reverse, so objects “on top” get found first
- Tables can be hidden
- Database Pre- and Post-delete notifications
  - (Not in this seeding ROM)
  - Launcher sends Pre-, and DataMgr sends Post-
- FldSetMaxVisibleLines, to know the maximum size that the field can expand to show
  - (Not in this seeding ROM)

Database Bundling

- A database is “Bundled” with its application if:
  - Same creator ID
  - dmHdrAttrCopyPrevention is not set
  - Either dmHdrAttrBundle is set, or type is Overlay
- Bundled databases should get moved, beamed, copied, etc. along with the app
- User won’t be able to tell there were bundled databases
- Use for libraries, default data

Recyclable Databases

- Sometimes it is hard to find a way to delete a database, if it is open. (Particularly if you're trying to delete your own application!)
- As of 4.0, just set dmHdrAttrRecyclable, and the database will be deleted as soon as it has been completely closed
- System will also delete recyclable databases during reboot, along with any other database with type 'temp', as can be seen in file streams

Debug ROM Changes

- CtlSetLabel and FrmCopyLabel verify they're being called for the right object kind
- FrmCopyTitle checks that there's enough room
- Most Frm* routines check for title, label, control text overlaps with the next object
- ErrFatalDisplay will truncate after 64 bytes
- DmReleaseRecord verifies that it was busy
  - That is, that DmGetRecord was called, not Query

Existing Known Bugs in the Seed

- TsmGlue* for gcc doesn't work
- Windows Constructor 16-bit icons don't work, though bitmaps do
- Don't edit non-monochrome bitmaps in PRCs
- Readme says object code vanishes. It doesn't
- A few in POSE; seed cd has 3.0a8e16
  - Three minor bugs were fixed for the final 3.0a8
- Unix POSE isn't on the CD

Reminders

- All the APIs are subject to change
- (Some have already changed since the seeding CD was burned)
- When you get new ROMs, always rebuild your application with the new includes
- Test your application!


accelerate
collaborate
innovate
communicate
palm
Q&A


accelerate
collaborate
innovate
communicate
palm
PalmSource
Conference 2000